FWeather

Gabriel Sansigolo

Brazilian National Institute for Space Research (INPE)





Introduction

Food security relies on accurate agricultural monitoring that covers both area and yield estimates. In major producing nations such as Brazil, in 2024, the gross domestic product was 11.7 trillion BRL, and agribusiness contributed approximately 30% of this value.



Source: pexels

Introduction

Meteorological data is crucial for agricultural monitoring. Gridded data of this type refers to a set of variables, such as precipitation, surface temperature, and wind, arranged into regularly spaced grids across a delimited region.



Source: pexels

Introduction

Despite these advancements, existing meteorological data packages have been modeled for a specific type of operation, not sharing methods across different scientific fields, using unique processes and data formats. They typically target analysts with deep programming backgrounds.



Source: pexels

Objective

This work describes **FWeather**, an open-source Python solution for meteorological data time series analysis. It uses the SpatioTemporal Asset Catalog (STAC) to access databases of big data and build virtual data cubes, facilitating the retrieval of time series.



FWeather

Source: Author

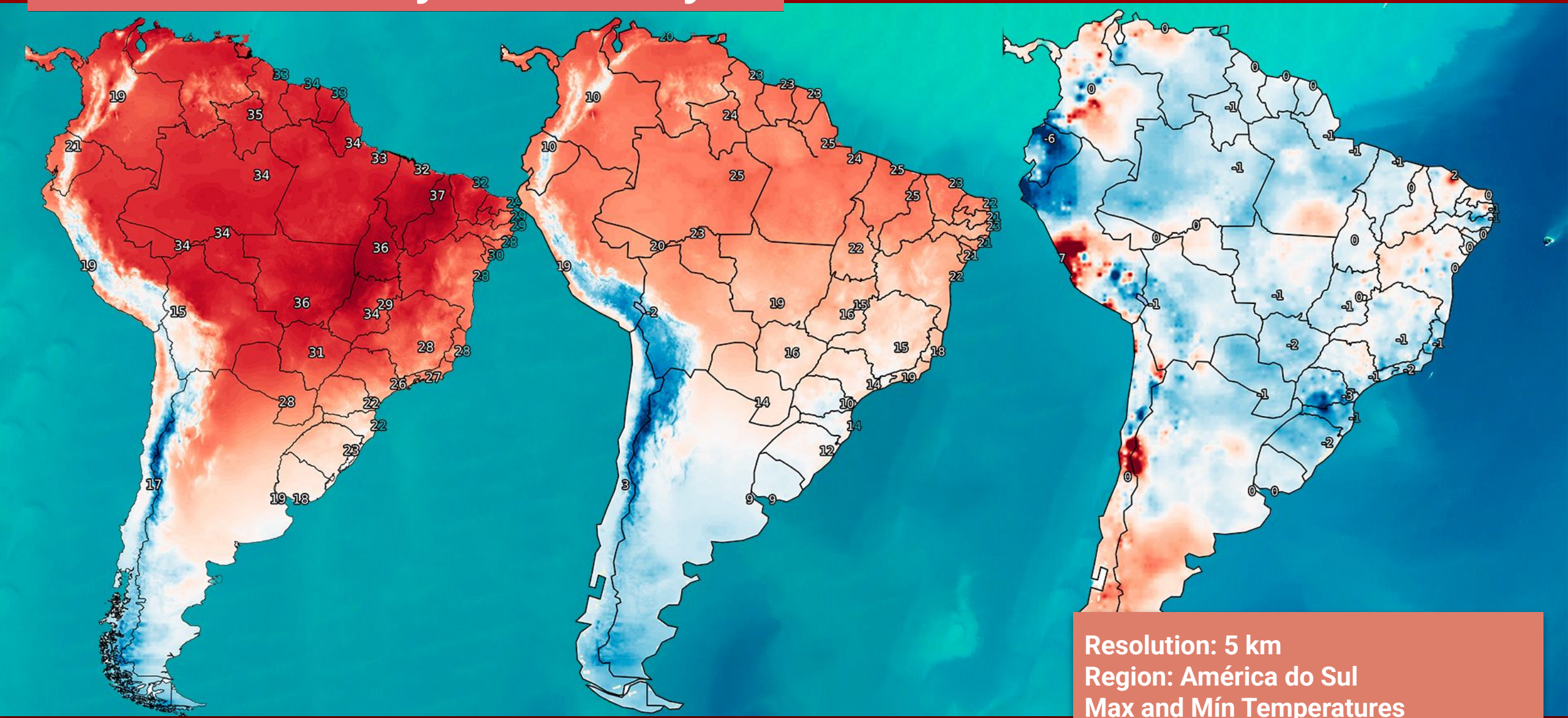
Meteorological Data

The term 'meteorological data' is widely used to refer to weather variables such as temperature, precipitation, and humidity, which serve a crucial role in a variety of applications, including Earth system modeling and agriculture.



Source: pexels

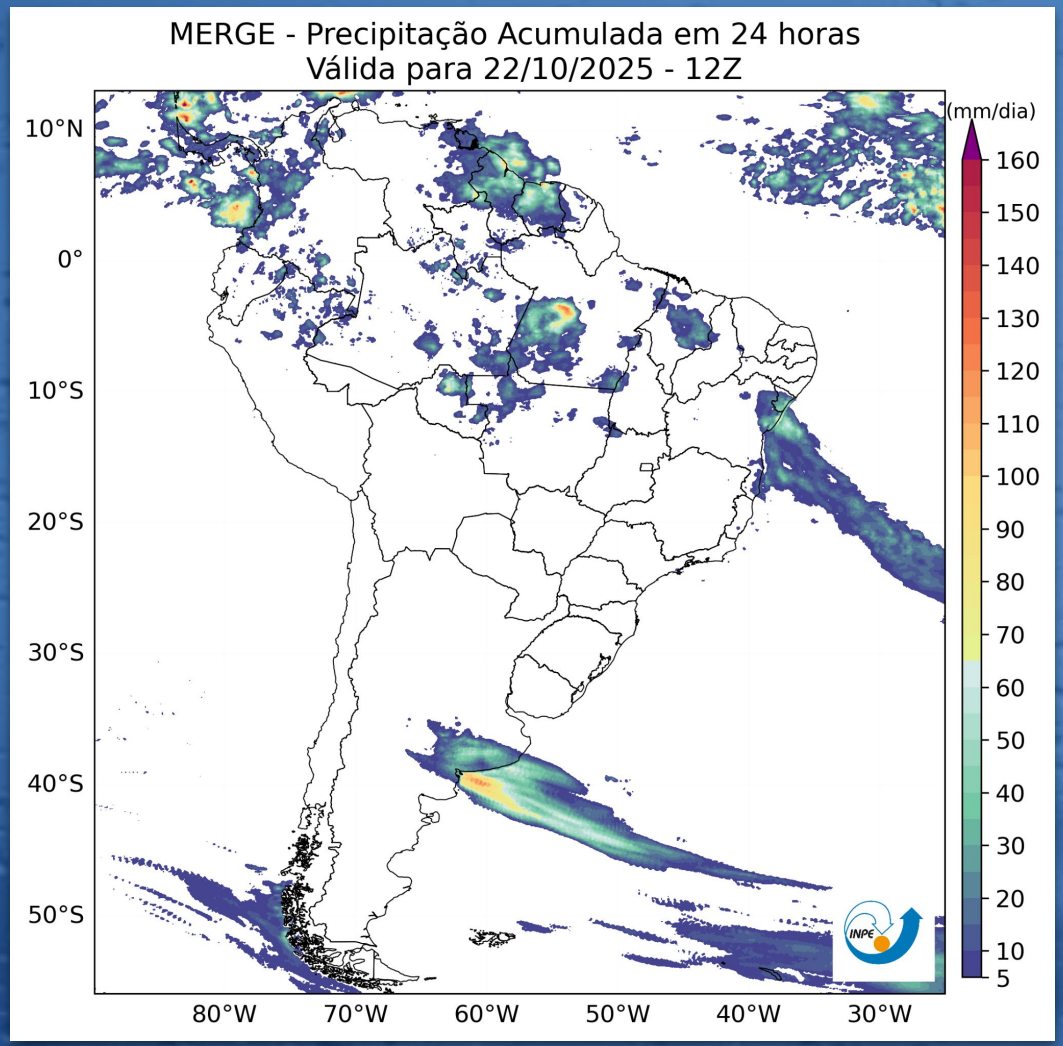
SAMeT / daily and hourly

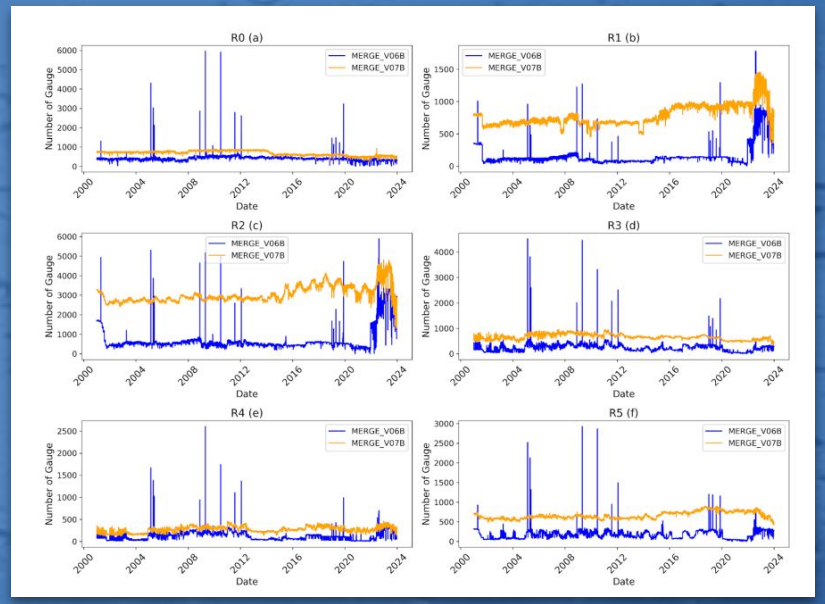


Resolution: 5 km
Region: América do Sul
Max and Mín Temperatures

Rozante, J. R., Ramirez, E., & Fernandes, A. d. A. (2022). A newly developed South American Mapping of Temperature with estimated lapse rate corrections. *International Journal of Climatology*, 42(4), 2135–2152. https://doi.org/10.1002/joc.7356

MERGE / daily and hourly





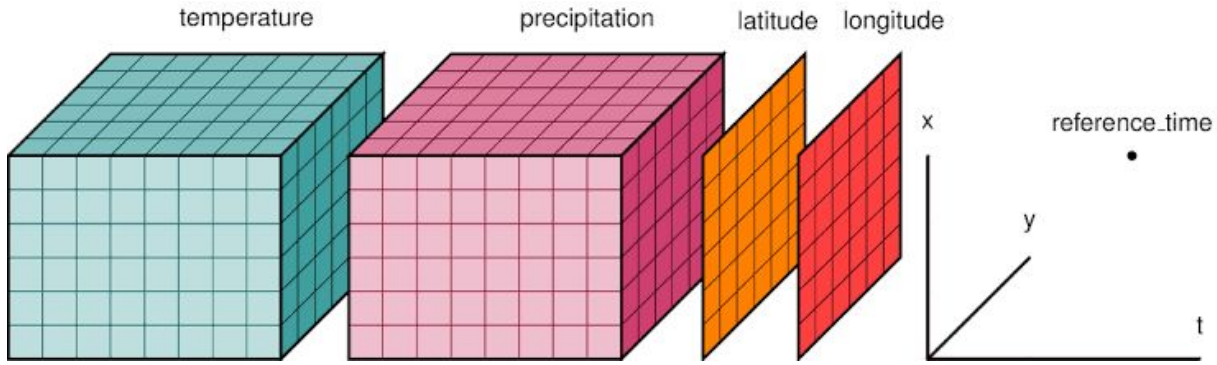
Resolution: 11 km
Region: América do Sul
Precipitation Levels

Rozante, *et.al* Combining TRMM and Surface Observations of Precipitation: Technique and Validation over South America
https://journals.ametsoc.org/view/journals/wefo/25/3/2010waf2222325_1.xml

FWeather

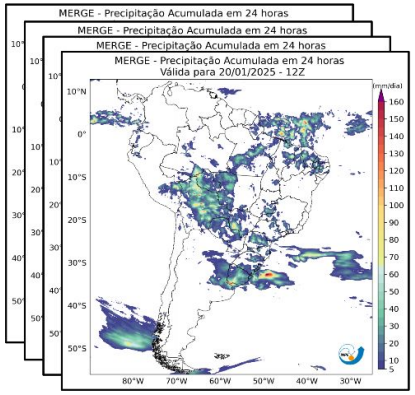
The tool provides methods for building multidimensional arrays and fetching time series of hourly precipitation, daily precipitation, temperature, and climate change projection data, all based on data cataloging and backed by computational infrastructure.

It allows large-scale agricultural monitoring using meteorological data time series, eliminating the need for massive local data storage.



Source: xarray

Architecture



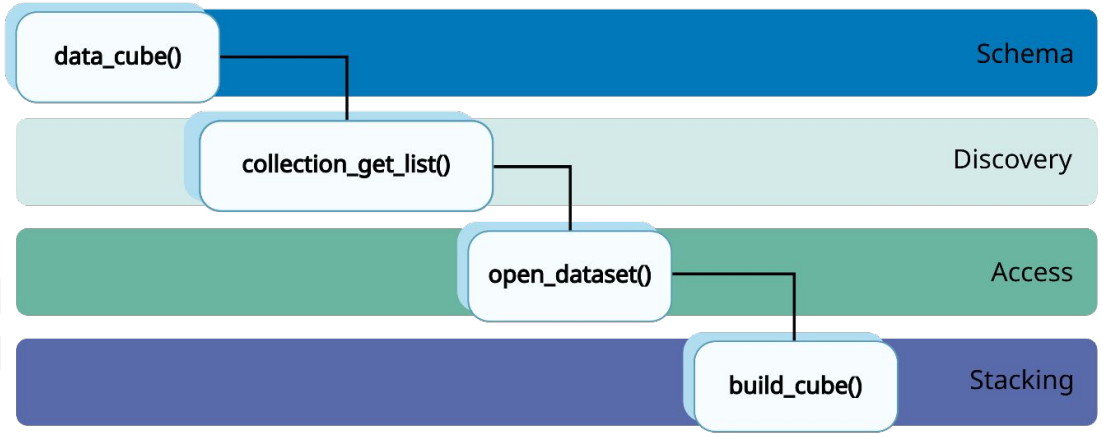
SAMeT Daily | SAMeT Hourly

MERGE Daily | MERGE Hourly

Eta Model

Python Client





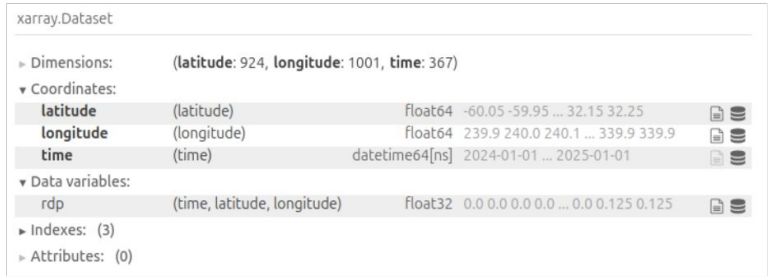


FWeather

{JSON}


```
{"values": [ 24.732938766479492,
 22.95409393310547,
 23.536516189575195,
 ...
 23.647729873657227],
"timeline": [ '2024-01-01T00:00:00',
'2024-01-02T00:00:00',
'2024-01-03T00:00:00',
...
```



Data cube function

```
from fweather import data_cube

stac_url="https://data.inpe.br/bdc/stac/v1"

prec_merge_cube = data_cube(
    stac_url=stac_url,
    collection="prec_merge_daily-1",
    start_date="2024-01-01",
    end_date="2024-12-31",
    bbox="-47.279,-17.072,-45.477,-15.448",
    bands=["merge_daily"]
)

print(prec_merge_cube)
```

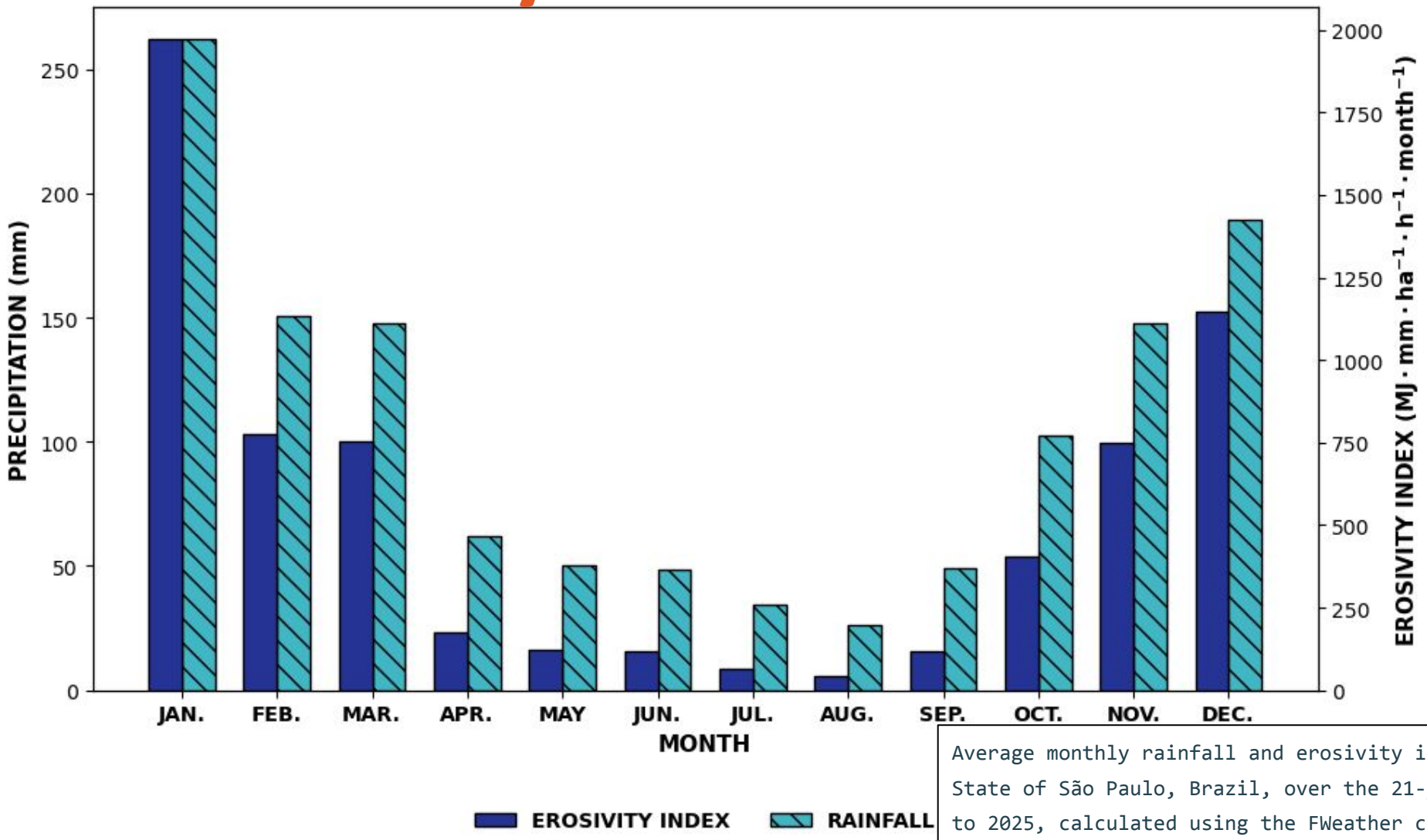
Rainfall erosivity mapping

Rainfall erosion is a natural process that occurs when raindrops strike the soil surface, promoting particle detachment and subsequent transport. The erosive potential of rainfall is quantified by the rainfall erosivity index (R-factor), which represents the climatic component of the Universal Soil Loss Equation (USLE).



Source: pexels

Rainfall erosivity



Average monthly rainfall and erosivity index for Campinas, State of São Paulo, Brazil, over the 21-year period from 2005 to 2025, calculated using the FWeather client.

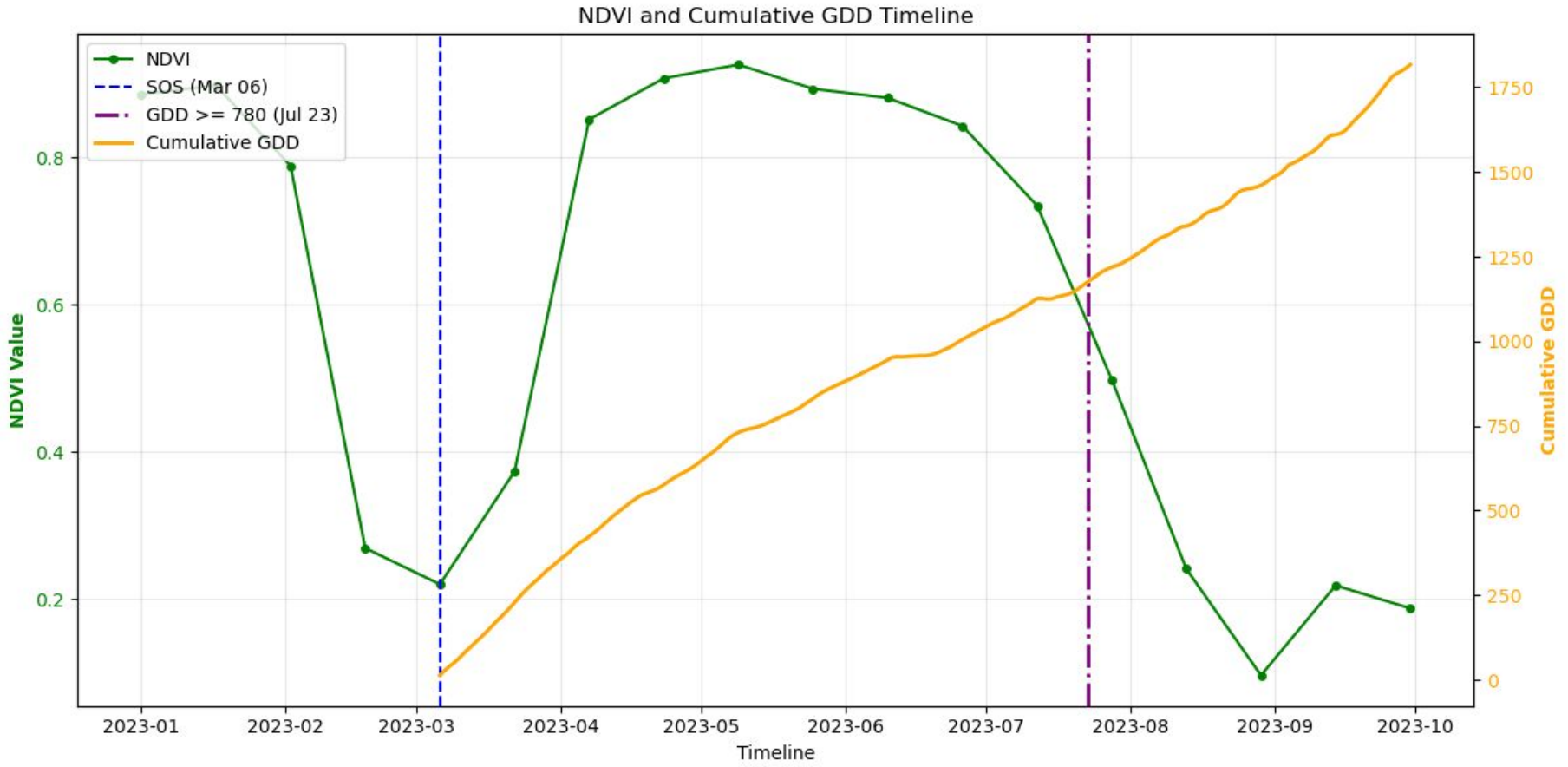
Growing degree days

Growing degree days (GDD) is a metric used to estimate plant growth based on temperature. It quantifies heat accumulation, which helps predict the development rates of plants throughout the growing season.



Source: pexels

GDD


NDVI and Cumulative GDD Timeline
NDVI
SOS (Mar 06)
GDD >= 780 (Jul 23)
Cumulative GDD
NDVI Value
Cumulative GDD
Timeline
0.2
0.4
0.6
0.8
0
250
500
750
1000
1250
1500
1750
2023-01
2023-02
2023-03
2023-04
2023-05
2023-06
2023-07
2023-08
2023-09
2023-10

Conclusion

The results of the rainfall erosivity map for the State of São Paulo using MERGE Daily precipitation and the GDD calculation for second-crop corn in Paraná using SAMeT Daily temperature show that we have successfully achieved the objective established: It was possible to perform large-scale monitoring using meteorological data time series without needing local data downloads.



FWeather

Source: Author

Code availability

PyPI FWeather:
https://pypi.org/project/fweather/

Github FWeather:
https://github.com/agrirslabinpe/fweather

Github Paper Compedium:
https://github.com/agrirslabinpe/meteorological-ts-paper

Experiments data:
https://doi.org/10.57760/sciencedb.43062

Read the Docs FWeather:
https://fweather.readthedocs.io/en/latest/

Thank you!

gabriel.sansigolo@inpe.br